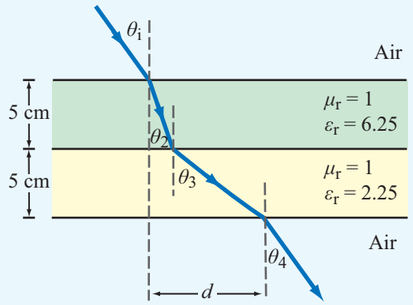


Figure P8.20 Problem P8.20.

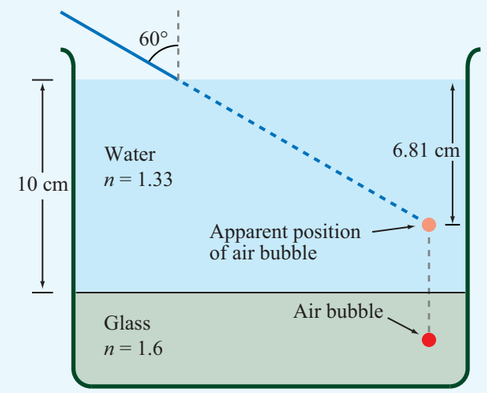


Figure P8.22 Apparent position of the air bubble in Problem 8.22.

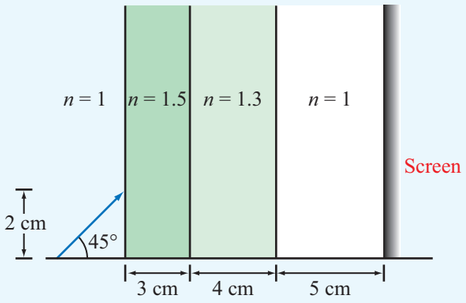


Figure P8.21 Light incident on a screen through a multilayered dielectric (Problem 8.21).

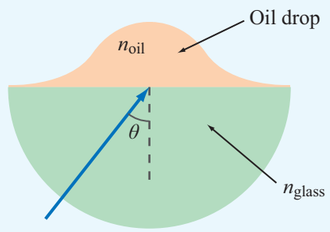


Figure P8.23 Oil drop on the flat surface of a glass semicylinder (Problem 8.23).

8.23 A glass semicylinder with $n = 1.5$ is positioned such that its flat face is horizontal, as shown in **Fig. P8.23**, and its horizontal surface supports a drop of oil, as also shown. When light is directed radially toward the oil, total internal reflection occurs if θ exceeds 53°. What is the index of refraction of the oil?

*__8.24__ A penny lies at the bottom of a water fountain at a depth of 30 cm. Determine the diameter of a piece of paper that when placed to float on the surface of the water directly above the penny, would totally obscure the penny from view. Treat the penny as a point and assume that $n = 1.33$ for water.

8.25 Suppose that the optical fiber of Example 8-5 is submerged in water (with $n = 1.33$) instead of air. Determine θ_a and f_p in that case.

*__8.26__ Equation (8.45) was derived for the case where the light incident upon the sending end of the optical fiber extends over the entire acceptance cone shown in **Fig. 8-12(b)**. Suppose the incident light is constrained to a narrower range extending between normal incidence and θ', where $\theta' < \theta_a$.

(a) Obtain an expression for the maximum data rate f_p in terms of θ'.

(b) Evaluate f_p for the fiber of Example 8-5 when $\theta' = 5°$.

Sections 8-4 and 8-5: Reflection and Transmission at Oblique Incidence

8.27 A 2-km long optical fiber uses a fiber core with $n_f = 1.6$ and a cladding with $n_c = 1.57$. Compute the maximum data

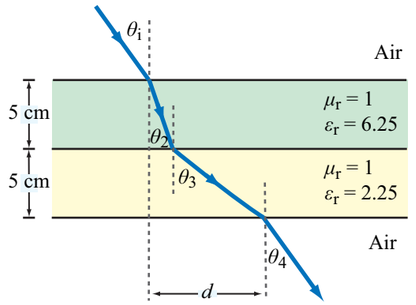


Figure P8.20 Problem P8.20.

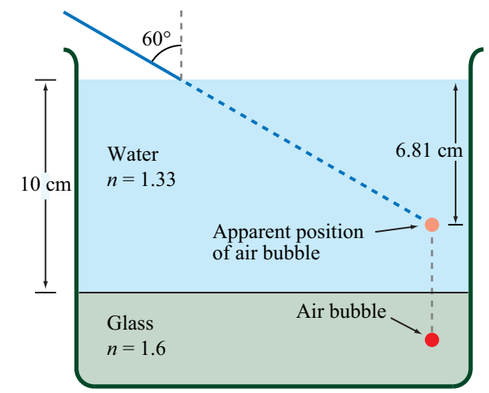


图 P8.22 问题 8.22 中气泡的明显位置。

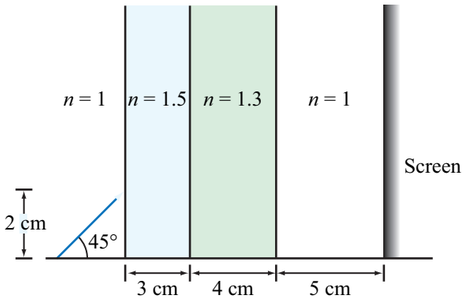


图 P8.21 光通过多层电介质入射到屏幕上（问题 8.21）。

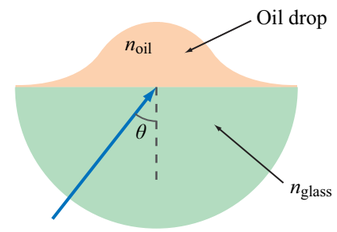


图 P8.23 玻璃半圆柱体平坦表面上的油滴（问题 8.23）。

8.23 将 n = 1.5 的玻璃半圆柱体放置在其平面水平的位置，如图 P8.23 所示，并且其水平表面支撑一滴油，也如图所示。当光径向射向油时，如果 θ 超过 53°，则会发生全内反射。油的折射率是多少？

*8.24 一枚硬币位于 30 厘米深的喷泉底部。确定一张纸的直径，当这张纸漂浮在便士正上方的水面上时，会完全遮住便士。将硬币视为一个点，并假设水的 n = 1.33。

8.25 假设例 8-5 中的光纤浸入水中（n = 1.33）而不是空气中。在这种情况下确定 θ_a 和 f_p。

*8.26 方程（8.45）是针对入射到光纤发送端的光延伸到图 8-12（b）所示的整个接收锥体的情况推导的。假设入射光被限制在法向入射和 θ′ 之间延伸的较窄范围内，其中 $\theta' < \theta_a$。

(a) 获得以 θ′ 表示的最大数据速率 f_p 的表达式。(b) 当 $\theta' = 5°$ 时，计算例 8-5 中光纤的 f_p。

第 8-4 节和第 8-5 节：斜入射时的反射和透射

8.27 2 公里长的光纤使用 $n_f = 1.6$ 的纤芯和 $n_c = 1.57$ 的包层。计算最大数据

rate and confirm your result using Module 8.2. The operating frequency is 100 THz.

8.28 A 3-km long optical fiber uses a fiber core with $n_f = 1.51$ and a cladding with $n_c = 1.48$. Compute the maximum data rate and confirm your result using Module 8.2. The operating frequency is 300 THz.

8.29 A plane wave in air with

$$\widetilde{\mathbf{E}}^i = \hat{\mathbf{y}}\, 20 e^{-j(3x+4z)} \qquad \text{(V/m)}$$

is incident upon the planar surface of a dielectric material with $\varepsilon_r = 4$, and it occupies the half-space $z \geq 0$. Determine:

(a) the polarization of the incident wave,

*(b) the angle of incidence,

(c) the time-domain expressions for the reflected electric and magnetic fields,

(d) the time-domain expressions for the transmitted electric and magnetic fields, and

(e) the average power density carried by the wave in the dielectric medium.

8.30 Repeat Problem 8.29 for a wave in air with

$$\widetilde{\mathbf{H}}^i = \hat{\mathbf{y}}\, 2 \times 10^{-2} e^{-j(8x+6z)} \qquad \text{(A/m)}$$

incident upon the planar boundary of a dielectric medium ($z \geq 0$) with $\varepsilon_r = 9$.

8.31 A plane wave in air with

$$\widetilde{\mathbf{E}}^i = (\hat{\mathbf{x}}\, 9 - \hat{\mathbf{y}}\, 4 - \hat{\mathbf{z}}\, 6) e^{-j(2x+3z)} \qquad \text{(V/m)}$$

is incident upon the planar surface of a dielectric material, with $\varepsilon_r = 2.25$, and it occupies the half-space $z \geq 0$. Determine:

*(a) the incidence angle θ_i,

(b) the frequency of the wave,

(c) the field $\widetilde{\mathbf{E}}^r$ of the reflected wave,

(d) the field $\widetilde{\mathbf{E}}^t$ of the wave transmitted into the dielectric medium, and

(e) the average power density carried by the wave into the dielectric medium.

8.32 Natural light is randomly polarized, which means that, on average, half the light energy is polarized along any given direction (in the plane orthogonal to the direction of propagation) and the other half of the energy is polarized along the direction orthogonal to the first polarization direction. Hence, when treating natural light incident upon a planar boundary, we can consider half of its energy to be in the form of parallel-polarized waves and the other half as perpendicularly polarized waves. Determine the fraction of the incident power reflected by the planar surface of a piece of glass with $n = 1.5$ when illuminated by natural light at 70°.

*8.33 A parallel-polarized plane wave is incident from air onto a dielectric medium with $\varepsilon_r = 9$ at the Brewster angle. What is the refraction angle?

8.34 Repeat Problem 8.33 using Module 8.3.

8.35 A perpendicularly polarized wave in air is obliquely incident upon a planar glass–air interface at an incidence angle of 30°. The wave frequency is 600 THz (1 THz = 10^{12} Hz), which corresponds to green light, and the index of refraction of the glass is 1.6. If the electric field amplitude of the incident wave is 50 V/m, determine:

(a) the reflection and transmission coefficients, and

(b) the instantaneous expressions for **E** and **H** in the glass medium.

8.36 Repeat part (a) of Problem 8.35 using Module 8.3.

8.37 Show that the reflection coefficient $\Gamma_\perp$ can be written in the following form:

$$\Gamma_\perp = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

8.38 Show that for nonmagnetic media, the reflection coefficient $\Gamma_\parallel$ can be written in the following form:

$$\Gamma_\parallel = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}.$$

*8.39 A parallel-polarized beam of light with an electric field amplitude of 10 (V/m) is incident in air on polystyrene with $\mu_r = 1$ and $\varepsilon_r = 2.6$. If the incidence angle at the air–polystyrene planar boundary is 50°, determine:

(a) the reflectivity and transmissivity, and

(b) the power carried by the incident, reflected, and transmitted beams if the spot on the boundary illuminated by the incident beam is 1 m^2 in area.

8.40 Repeat part (a) of Problem 8.39 using Module 8.3.

8.41 A 50-MHz right-hand circularly polarized plane wave with an electric field modulus of 30 V/m is normally incident in air upon a dielectric medium with $\varepsilon_r = 9$ and occupying the region defined by $z \geq 0$.

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$.

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使用模块 8.2 评分并确认您的结果。工作频率为100太赫兹。

8.28 3 公里长的光纤使用 $n_f = 1.51$ 的纤芯和 $n_c = 1.48$ 的包层。使用模块 8.2 计算最大数据速率并确认您的结果。工作频率为300太赫兹。

8.29 空气中的平面波

$$\widetilde{\mathbf{E}}^i = \hat{\mathbf{y}}\, 20 e^{-j(3x+4z)} \qquad \text{(V/m)}$$

入射到介电材料的平面上，$\varepsilon_r = 4$，它占据半空间 $z \geq 0$。确定：

(a) 入射波的偏振，

*(b) 入射角，

(c) 反射电场和磁场的时域表达式，(d) 透射电场和磁场的时域表达式，以及 (e) 介电介质中波所携带的平均功率密度。

8.30 对于空气中的波，重复问题 8.29

$$\widetilde{\mathbf{H}}^i = \hat{\mathbf{y}}\, 2 \times 10^{-2} e^{-j(8x+6z)} \qquad \text{(A/m)}$$

入射到 $\varepsilon_r = 9$ 的介电介质 ($z \geq 0$) 的平面边界上。

8.31 空气中的平面波

$$\widetilde{\mathbf{E}}^i = (\hat{\mathbf{x}}\, 9 - \hat{\mathbf{y}}\, 4 - \hat{\mathbf{z}}\, 6) e^{-j(2x+3z)} \qquad \text{(V/m)}$$

入射到介电材料的平面上，$\varepsilon_r = 2.25$，它占据半空间 $z \geq 0$。确定：

*(a) 入射角 θ_i，

(b) 波的频率，

(c) 反射波的场 $\widetilde{E}^r$，

(d) 传输到介电介质中的波的场 $\widetilde{E}^t$，以及 (e) 进入介电介质的波所携带的平均功率密度。

8.32 自然光是随机偏振的，这意味着平均而言，一半的光能沿任何给定方向（在与传播方向正交的平面内）偏振，另一半能量沿与第一个偏振方向正交的方向偏振。因此，当处理入射到平面边界上的自然光时，我们可以认为其能量的一半为平行偏振波，另一半为垂直偏振波。

波浪。确定在 70° 的自然光照射下，$n = 1.5$ 的玻璃平面反射的入射功率的分数。

*8.33 平行偏振平面波从空气以布儒斯特角入射到$\varepsilon_r = 9$ 的介电介质上。什么是折射角？

8.34 使用模块 8.3 重复问题 8.33。

8.35 空气中的垂直偏振波以 30° 的入射角倾斜入射到平面玻璃-空气界面上。波频率为600 THz（1 THz = 10^{12} Hz），对应于绿光，玻璃的折射率为1.6。如果入射波的电场幅度为50 V/m，则确定：

(a) 反射系数和透射系数，以及 (b) 玻璃介质中 E 和 H 的瞬时表达式。

8.36 使用模块 8.3 重复问题 8.35 的 (a) 部分。

8.37 证明反射系数$\Gamma_\perp$可以写成以下形式：

$$\Gamma_\perp = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

8.38 证明对于非磁性介质，反射系数 $\Gamma_\parallel$ 可以写成以下形式：

$$\Gamma_\parallel = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}.$$

*8.39 电场强度为 10 (V/m) 的平行偏振光束在空气中入射到聚苯乙烯上，$\mu_r = 1$ 和 $\varepsilon_r = 2.6$。如果空气-聚苯乙烯平面边界处的入射角为 50°，则确定：

(a) 反射率和透射率，以及 (b) 如果入射光束照射的边界上的点面积为 1 m^2，则入射光束、反射光束和透射光束携带的功率。

8.40 使用模块 8.3 重复问题 8.39 的 (a) 部分。

8.41 电场模量为 30 V/m 的 50 MHz 右旋圆极化平面波通常在空气中入射到 $\varepsilon_r = 9$ 的介电介质上，并占据由 $z \geq 0$ 定义的区域。

(a) 假设电场在 $z = 0$ 和 $t = 0$ 处为正最大值，请写出入射波电场相量的表达式。

(b) Calculate the reflection and transmission coefficients.

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted wave, and the total field in the region $z \leq 0$.

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium.

8.42 Consider a flat 5-mm thick slab of glass with $\varepsilon_r = 2.56$.

*(a) If a beam of green light ($\lambda_0 = 0.52\ \mu$m) is normally incident upon one of the sides of the slab, what percentage of the incident power is reflected back by the glass?

(b) To eliminate reflections, it is desired to add a thin layer of antireflection coating material on each side of the glass. If you are at liberty to specify the thickness of the antireflection material as well as its relative permittivity, what would these specifications be?

8.43 A perpendicular polarization wave in air is incident upon the surface of a perfect conductor at an incidence angle of 30°. The frequency of the wave is 2 GHz. Use Module 8.1 to determine the location of the nearest electric field maximum from the conductor surface.

8.44 A 2 GHz plane wave in air is incident upon a conducting medium with $\varepsilon_r = 4$ and $\sigma = 2$ S/m at an incidence angle of 60°. The wave is parallel-polarized. Use Module 8.4 to compute:

(a) the reflectivity and transmissivity, and

(b) the effective refraction angle.

8.45 A 2 GHz plane wave in air is incident upon a conducting medium with $\varepsilon_r = 4$ and $\sigma = 2$ S/m at an incidence angle of 60°. The wave is perpendicularly polarized. Use Module 8.4 to compute:

(a) the reflectivity and transmissivity, and

(b) the effective refraction angle.

Sections 8-6 to 8-11: Waveguides and Resonators

8.46 Derive Eq. (8.89b).

*8.47 A hollow, rectangular waveguide is to be used to transmit signals at a carrier frequency of 6 GHz. Choose its dimensions so that the cutoff frequency of the dominant TE mode is lower than the carrier by 25% and that of the next mode is at least 25% higher than the carrier.

8.48 A TE wave propagating in a dielectric-filled waveguide of unknown permittivity has dimensions $a = 5$ cm and $b = 3$ cm. If the x component of its electric field is given by

$$E_x = -36\cos(40\pi x)\sin(100\pi y) \cdot \sin(2.4\pi \times 10^{10} t - 52.9\pi z), \quad \text{(V/m)}$$

determine:

(a) the mode number,

(b) ε_r of the material in the guide,

(c) the cutoff frequency, and

(d) the expression for H_y.

*8.49 A waveguide filled with a material whose $\varepsilon_r = 2.25$ has dimensions $a = 2$ cm and $b = 1.4$ cm. If the guide is to transmit 10.5 GHz signals, what possible modes can be used for the transmission?

8.50 For a rectangular waveguide operating in the TE_{10} mode, obtain expressions for the surface charge density $\tilde{\rho}_s$ and surface current density $\tilde{\mathbf{J}}_s$ on each of the four walls of the guide.

*8.51 A waveguide with dimensions $a = 1$ cm and $b = 0.7$ cm is to be used at 20 GHz. Determine the wave impedance for the dominant mode when

(a) the guide is empty, and

(b) the guide is filled with polyethylene (whose $\varepsilon_r = 2.25$).

8.52 A narrow rectangular pulse superimposed on a carrier with a frequency of 9.5 GHz was used to excite all possible modes in a hollow guide with $a = 3$ cm and $b = 2.0$ cm. If the guide is 100 m in length, how long will it take each of the excited modes to arrive at the receiving end?

*8.53 If the zigzag angle θ' is 25° for the TE_{10} mode, what would it be for the TE_{20} mode?

8.54 Measurement of the TE_{101} frequency response of an air-filled cubic cavity revealed that its Q is 4802. If its volume is 64 mm^3, what material are its sides made of? [*Hint:* See Appendix B.]

8.55 A hollow cavity made of aluminum has dimensions $a = 4$ cm and $d = 3$ cm. Calculate Q of the TE_{101} mode for

*(a) $b = 2$ cm, and

(b) $b = 3$ cm.

8.56 A hollow rectangular waveguide with dimensions $a = 2$ cm and $b = 1$ cm is used to propagate signals at 20 GHz. Use Module 8.5 to determine:

(a) the cutoff frequency of TE_{10} mode, and

(b) all possible transmission modes.

8.57 A hollow rectangular waveguide with dimensions $a = 4$ cm and $b = 2$ cm is used to propagate signals at 12 GHz. Use Module 8.5 to determine:

(a) the cutoff frequency of TE_{10} mode, and

(b) all possible transmission modes.

(b) 计算反射和透射系数。 (c) 写出反射波、透射波和 z ≤ 0 区域中总场的电场相量表达式。 (d) 确定边界反射并传输到第二介质中的入射平均功率的百分比。 8.42 考虑一块 5 毫米厚的玻璃板，其 ε_r = 为 2.56。

*(a) 如果一束绿光 ($\lambda_0 = 0.52\ \mu$m) 垂直入射到平板的一侧，则玻璃反射回来的入射功率的百分比是多少？

(b) 为了消除反射，需要在玻璃的每一面添加一层薄薄的减反射涂层材料。如果您可以自由指定抗反射材料的厚度及其相对介电常数，那么这些规格是什么？

8.43 空气中的垂直极化波以 30° 的入射角入射到理想导体的表面。波的频率为 2 GHz。使用模块 8.1 确定距导体表面最近的电场最大值的位置。

8.44 空气中的 2 GHz 平面波以 60° 的入射角入射到 $\varepsilon_r = 4$ 和 σ = 2 S/m 的导电介质上。该波是平行偏振的。使用模块 8.4 计算：

(a) 反射率和透射率，以及
(b) 有效折射角。

8.45 空气中的 2 GHz 平面波以 60° 的入射角入射到 $\varepsilon_r = 4$ 和 σ = 2 S/m 的导电介质上。波是垂直偏振的。使用模块 8.4 计算：

(a) 反射率和透射率，以及 (b) 有效折射角。

第 8-6 节到 8-11 节：波导和谐振器

8.46 推导方程(8.89b)。

*8.47 空心矩形波导用于传输载波频率为 6 GHz 的信号。选择其尺寸，使主 TE 模式的截止频率比载波低 25%，而下一个模式的截止频率至少比载波高 25%。

8.48 在介电常数未知的电介质填充波导中传播的 TE 波的尺寸为 a = 5 cm 和 b = 3 cm。如果其电场的 x 分量由下式给出

$$E_x = -36\cos(40\pi x)\sin(100\pi y) \cdot \sin(2.4\pi \times 10^{10} t - 52.9\pi z), \quad \text{(V/m)}$$

决定：
(a) 模数，(b) 波导中材料的 ε_r，
(c) 截止频率，以及 (d) H_y 的表达式。

*8.49 填充有 $\varepsilon_r = 2.25$ 材料的波导，其尺寸为 a = 2 cm 和 b = 1.4 cm。如果要传输10.5GHz信号，可以采用哪些模式来传输？

8.50 对于工作在 TE_{10} 模式的矩形波导，求得波导四个壁上的表面电荷密度 $\tilde{\rho}_s$ 和表面电流密度 $\tilde{J}_s$ 的表达式。

*8.51 在 20 GHz 下使用尺寸 a = 1 cm 和 b = 0.7 cm 的波导。确定主模式的波阻抗

(a) 指南是空的，并且
(b) 导向器填充有聚乙烯（其$\varepsilon_r = 2.25$）。

8.52 使用叠加在频率为 9.5 GHz 的载波上的窄矩形脉冲来激励 a = 3 cm 和 b = 2.0 cm 的空心波导中的所有可能模式。如果波导长度为 100 m，则每个激发模式需要多长时间才能到达接收端？

*8.53 如果锯齿角θ′是 25°对于TE_{10}模式，对于 TE 来说会是什么？$_{20}$模式？

8.54 对充气立方体空腔的 TE_{101} 频率响应的测量表明其 Q 为 4802。如果其体积为 64 mm^3，其侧面由什么材料制成？ [提示：请参阅附录 B。]

8.55 铝制空腔的尺寸 a = 4 cm，d = 3 cm。计算 TE_{101} 模式的 Q

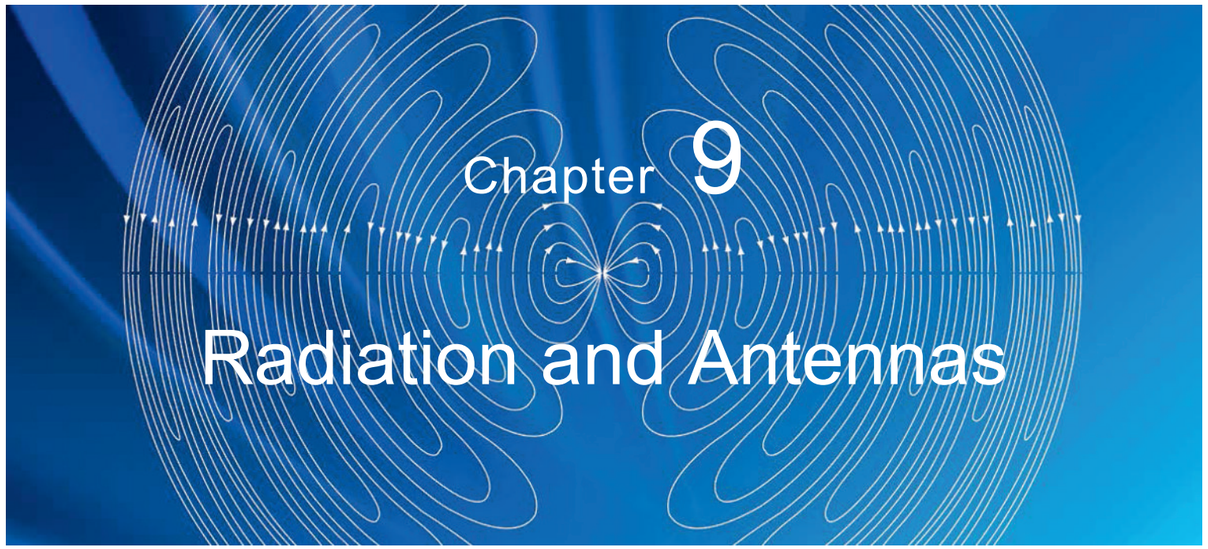
*(a)b=2厘米，和(
b)b=3厘米。

8.56 使用尺寸 a = 2 cm 和 b = 1 cm 的空心矩形波导来传播 20 GHz 的信号。使用模块 8.5 确定：

(a) TE_{10} 模式的截止频率，以及 (b) 所有可能的传输模式。

8.57 使用尺寸 a = 4 cm 和 b = 2 cm 的空心矩形波导来传播 12 GHz 信号。使用模块 8.5 确定：

(a) TE_{10} 模式的截止频率，以及 (b) 所有可能的传输模式。



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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Calculate the electric and magnetic fields of waves radiated by a dipole antenna.
2. Characterize the radiation of an antenna in terms of its radiation pattern, directivity, beamwidth, and radiation resistance.
3. Apply the Friis transmission formula to a free-space communication system.
4. Calculate the electric and magnetic fields of waves radiated by aperture antennas.
5. Calculate the radiation pattern of multi-element antenna arrays.

第9章

辐射与天线

章节内容

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目标

学习本章介绍的材料后，您应该能够：

1. 计算偶极天线辐射的电场和磁场。 2. 通过辐射方向图、方向性、波束宽度和辐射电阻来表征天线的辐射。 3. 将 Friis 传输公式应用于自由空间通信系统。 4. 计算孔径天线辐射波的电场和磁场。 5. 计算多元件天线阵列的辐射方向图。

Overview

An ***antenna*** is a transducer that converts a guided wave propagating on a transmission line into an electromagnetic wave propagating in an unbounded medium (usually free space), or vice versa. **Figure 9-1** shows how a wave is launched by a hornlike antenna with the horn acting as the transition segment between the waveguide and free space.

Antennas are made in various shapes and sizes (**Fig. 9-2**) and are used in radio and television broadcasting and reception, radio-wave communication systems, cellular telephones, radar systems, anticollision automobile sensors, and many other applications. The radiation and impedance properties of an antenna are governed by its shape, size, and material properties. The dimensions of an antenna are usually measured in units of λ of the wave it is launching or receiving; a 1-m long dipole antenna operating at a wavelength $\lambda = 2$ m exhibits the same properties as a 1-cm long dipole operating at $\lambda = 2$ cm. Hence, in most of our discussions in this chapter, we refer to antenna dimensions in wavelength units.

Reciprocity

The directional function characterizing the *relative* distribution of power radiated by an antenna is known as the ***antenna radiation pattern*** (or simply the ***antenna pattern***). An ***isotropic*** antenna is a hypothetical antenna that radiates equally in all directions, and it is often used as a reference radiator when describing the radiation properties of real antennas.

► Most antennas are ***reciprocal*** devices, exhibiting the same radiation pattern for transmission as for reception. ◄

Reciprocity means that, if in the transmission mode a given antenna transmits in direction A 100 times the power it transmits in direction B, then when used in the reception mode it is 100 times more sensitive to electromagnetic radiation incident from direction A than from B. All of the antennas shown in **Fig. 9-2** obey the reciprocity law, but not all antennas are reciprocal devices. Reciprocity may not hold for some solid-state antennas composed of nonlinear semiconductors or ferrite materials. Such nonreciprocal antennas are beyond the scope of this chapter; hence, reciprocity is assumed throughout. The reciprocity property is very convenient because it allows us to compute the radiation pattern of an antenna in the transmission mode—even when the antenna is intended to operate as a receiver.

To fully characterize an antenna, one needs to study its radiation properties and impedance. The radiation properties include its directional radiation pattern and the associated polarization state of the radiated wave when the antenna is used in the transmission mode, which is also called the ***antenna polarization***.

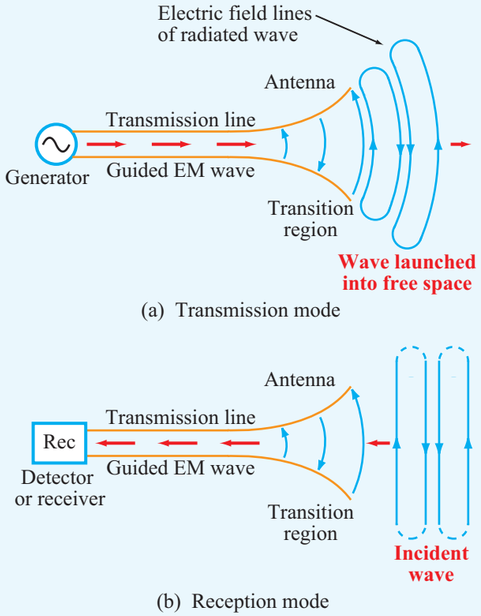


Figure 9-1 Antenna as a transducer between a guided electromagnetic wave and a free-space wave, for both transmission and reception.

► Being a reciprocal device, an antenna, when operating in the receiving mode, can extract from an incident wave only that component of the wave whose electric field matches the antenna polarization state. ◄

The second aspect, the ***antenna impedance***, pertains to the transfer of power from a generator to the antenna when the antenna is used as a transmitter and, conversely, the transfer of power from the antenna to a load when the antenna is used as a receiver, as will be discussed later in Section 9-5. It should be noted that throughout our discussions in this chapter it will be assumed that the antenna is properly matched to the transmission line connected to its terminals, thereby avoiding reflections and their associated problems.

概述

天线是一种换能器，可将传输线上传播的导波转换为在无界介质（通常是自由空间）中传播的电磁波，反之亦然。图 9-1 显示了如何通过喇叭状天线发射波，其中喇叭充当波导和自由空间之间的过渡段。

天线有各种形状和尺寸（图9-2），用于无线电和电视广播和接收、无线电波通信系统、蜂窝电话、雷达系统、防碰撞汽车传感器和许多其他应用。天线的辐射和阻抗特性由其形状、尺寸和材料特性决定。天线的尺寸通常以其发射或接收的波的 λ 为单位进行测量；在波长 $\lambda = 2$ m 下工作的 1 米长偶极子天线表现出与在 $\lambda = 2$ cm 下工作的 1 厘米长偶极子天线相同的特性。因此，在本章的大部分讨论中，我们提到的天线尺寸都是以波长为单位的。

互易

表征天线辐射功率相对分布的方向函数称为天线辐射方向图（或简称为天线方向图）。各向同性天线是一种在所有方向上均匀辐射的假想天线，在描述实际天线的辐射特性时通常将其用作参考辐射器。

► 大多数天线都是互易设备，在发射和接收时表现出相同的辐射方向图。 ◄

互易性是指，如果在发射模式下，给定天线在 A 方向上发射的功率是在 B 方向上发射的功率的 100 倍，那么当用于接收模式时，它对从 A 方向入射的电磁辐射的敏感度比从 B 方向入射的电磁辐射敏感 100 倍。图 9-2 中所示的所有天线都遵循互易定律，但并非所有天线都是互易器件。对于某些由非线性半导体或铁氧体材料组成的固态天线，互易性可能不成立。这种非互易天线超出了本章的范围；因此，自始至终都假定互易。互易性非常方便，因为它允许我们计算传输模式下天线的辐射方向图——即使天线打算用作接收器。

为了全面表征天线，需要研究其辐射特性和阻抗。辐射特性包括天线的定向辐射图和辐射波的相关极化状态。

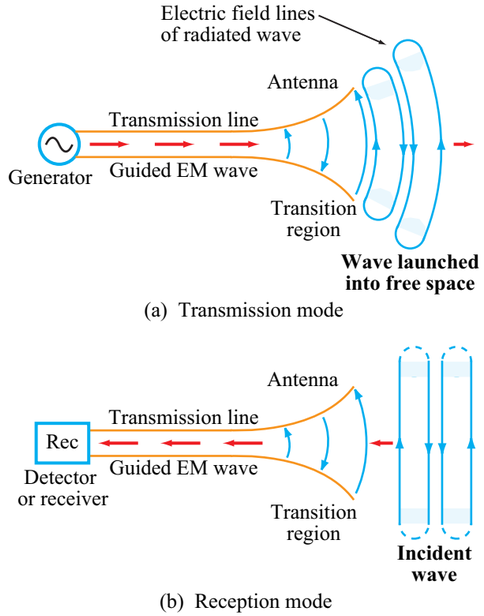


图 9-1 天线作为引导电磁波和自由空间波之间的换能器，用于传输和接收。

用于传输模式，也称为天线极化。

► 作为一种互易装置，天线在接收模式下工作时，只能从入射波中提取电场与天线极化状态匹配的波分量。 ◄

第二个方面，即天线阻抗，涉及当天线用作发射器时从发电机到天线的功率传输，以及相反，当天线用作接收器时从天线到负载的功率传输，稍后将在第 9-5 节中讨论。应该注意的是，在本章的整个讨论中，我们都假设天线与连接到其端子的传输线正确匹配，从而避免反射及其相关问题。